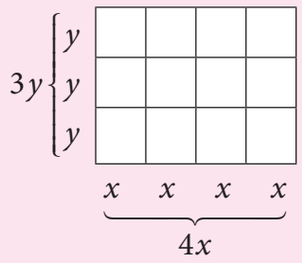
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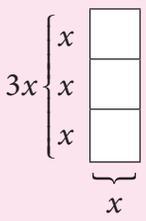
Try these

1. Observe the following figures and try to find its area, geometrically. Also verify the same by multiplication of monomial.

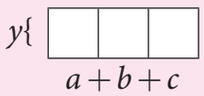
i)



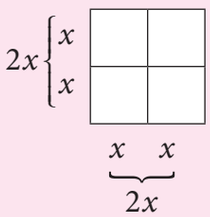
ii)



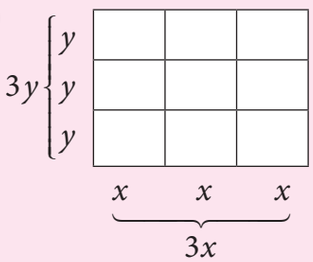
iii)



iv)



ii)



2. Let the length and breadth of a tile be x and y respectively. Using such tiles construct as many rectangles as you can and find out the length and breadth of the rectangles so formed such that its area is

(i) $12xy$ (ii) $8xy$ (iii) $9xy$

3.3 Geometrical proof of Identities

By using this concept of multiplication of monomials, let us try to prove the identities geometrically, which are very much useful in solving algebraic problems.

3.3.1 Identity-1: $(x+a)(x+b) = x^2 + x(a+b) + ab$

Consider four regions. One region is square shaped with dimension 3×3 (Grey). Also, the other three regions are rectangle in shape with dimensions 4×3 (yellow), 3×2 (red) and 4×2 (Blue).

Arrange these four regions to form a rectangular shape as shown in the *Fig. 3.8*.

By observing *Fig. 3.8*, we can note that,

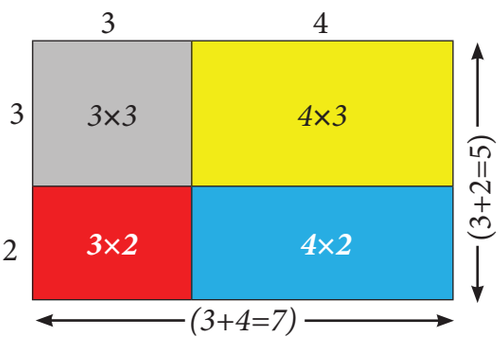


Fig. 3.8

Area of the bigger rectangle = Area of a square (Grey) + Area of Three rectangles

$$(3+4)(3+2) = (3\times3) + (4\times3) + (3\times2) + (4\times2)$$

$$(3+4)(3+2) = (3\times3) + 3\times(4+2) + (4\times2) \quad \ldots(1)$$

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